

Early Detection of Silent Machine Learning Pipeline Failure

Through Operational Monitoring

A Marketplace Analytics Case Study

Master of Science / Informatik

IU Internationale Hochschule

Berlin, Germany

Student Name: Abhishek Prasad

Student ID: 102301095

Submission Date: 01.10.2026

Table of Contents

Table of Contents	2
Acknowledgement	5
Abstract.....	5
List of Figures.....	6
List of Abbreviations	6
Chapter 1: Introduction	8
1.1 Background	8
1.2 Problem Statement.....	9
1.3 Research Aim	9
1.4 Research Objectives	9
1.5 Research Questions	10
1.6 Research Gap	10
1.7 Structure of Research	11
1.8 Chapter Summary	12
Chapter 2: Literature Review	13
2.1 Introduction	13
2.2 Technical Debt in Machine Learning Systems	13
2.3 Concept Drift and Data Drift	14
2.4 Monitoring Practice for Deployed Models.....	14
2.5 Forecast Accuracy Measurement	15
2.6 Gradient Boosting for Tabular Prediction.....	16
2.7 Dimensional Modelling and the Semantic Layer.....	16
2.8 Synthetic Data as an Evaluation Instrument	17
2.9 Research Gap Identified in the Literature.....	18
Chapter 3: Methodology.....	19
3.1 Research Design	19
3.2 Data Generation	19
3.2.1 Structural realism	19
3.2.2 Behavioural modelling.....	19
3.2.3 Consistency by construction	20
3.3 Warehouse and Semantic Layer Construction.....	20
3.4 Model Development	21
3.4.1 Demand forecasting	21
3.4.2 Technician ranking	21
3.4.3 Dynamic pricing	22
3.5 Failure Injection and Measurement Protocol.....	22
3.6 Validation Strategy	22
3.7 Ethical Considerations.....	23
3.8 Chapter Summary	23
Chapter 4: Discussion and Analysis	25

4.1 Introduction	25
4.2 Overview of the Dataset	25
4.3 Findings	25
Theme 1: Detection latency differs by two and a half months between signal classes	25
Theme 2: A stale model degrades only when the environment changes	26
Theme 3: MAPE conceals a failure mode it cannot represent	26
Theme 4: Model quality assessed honestly, including where it is poor	27
Theme 5: Automated verification detected defects that review did not	28
Supplementary findings	28
4.4 Interpretation of Findings	29
4.4.1 The significance of outcome independence	29
4.4.2 Accuracy as a lagging indicator	29
4.4.3 Metric selection as a design decision	29
4.4.4 Honest reporting as a methodological requirement	30
4.4.5 Verification layers as independent instruments	30
4.5 Discussion of Findings	30
4.5.1 Relation to the technical debt literature	30
4.5.2 Relation to drift detection literature	30
4.5.3 Relation to forecast evaluation literature	31
4.5.4 Implications for practice	31
4.6 Limitations of Findings	31
4.6.1 Synthetic data	31
4.6.2 Single failure mode	32
4.6.3 Model evaluation constraints	32
4.6.4 Single-domain scope	32
4.7 Chapter Summary	32
Chapter 5: Conclusion and Recommendations	33
5.1 Introduction	33
5.2 Summary of Key Findings	33
5.3 Limitations of Study	33
5.4 Link to Research Questions and Objectives	33
5.5 Recommendations	34
5.5.1 For practice	34
5.5.2 For further research	35
5.6 Reflections in First Person	35
References	37
Appendix 1: Declaration of Authenticity	39
Appendix 2: Artefact Inventory	40
Appendix 3: Validation Output	41
Declaration of Authenticity	42

To populate this table in Word: select it, then press F9 and choose "Update entire table".

Acknowledgement

I would like to express my gratitude to my supervisor for their guidance throughout this project, and to the authors whose published work on machine learning system reliability, dimensional modelling and forecast evaluation provided the theoretical foundation for this study. I also thank my family and friends for their encouragement during its completion.

Abstract

Machine learning systems fail differently from conventional software. A web service that breaks stops responding; a model that breaks continues to return predictions, and those predictions remain plausible long after the system that produces them has stopped being maintained. This project investigates whether operational monitoring signals can detect such a silent failure earlier than the accuracy metrics conventionally used to supervise deployed models.

An end-to-end analytics and machine learning platform was constructed for a simulated home-services marketplace operating in Bengaluru, India, comprising 57,973 bookings, 24,000 customers and 850 service professionals across twenty months. The platform includes a dimensional data warehouse of eleven tables, a semantic layer of 119 measures, two dashboard implementations, three trained machine learning services, and a monitoring layer covering eight models.

A pipeline failure was deliberately injected at a known date: the scheduled retraining job for the demand forecasting model ceases to succeed on 15 March

1. Detection performance was then measured for two classes of signal —

accuracy metrics and operational metrics — against that known ground truth.

The principal finding is that training-data age, an operational metric requiring no model evaluation and no labelled outcomes, diverged from its normal pattern on the day of failure, whereas the model's accuracy metric showed no degradation for seventy-eight days. Accuracy only deteriorated when an exogenous seasonal regime change exposed the stale model. A secondary finding concerns metric selection: the mean absolute percentage error of the forecasting model was 10.4%, while its weighted absolute percentage error over the same predictions was 37.2%. The divergence is attributable to 11,004 forecast cells in which

demand was predicted and none occurred, a failure mode that MAPE is mathematically incapable of observing.

The study concludes that model monitoring should include at least one signal that does not depend on outcome labels, and that percentage-error metrics should not be reported without a companion metric that accounts for zero-actual cases.

Keywords: machine learning operations, model monitoring, concept drift, data drift, dimensional modelling, demand forecasting, forecast evaluation, business intelligence, gradient boosting, marketplace analytics

List of Figures

Figure	Title
Figure 1	Star schema of the analytical model
Figure 2	Operations dashboard: service level against capacity strain
Figure 3	Capacity strain versus SLA breach, daily observations
Figure 4	Seasonal demand by service category
Figure 5	Technician utilisation by trade and season
Figure 6	Model health dashboard: the four-beat failure sequence
Figure 7	Training-data age, showing divergence from 15 March 2026
Figure 8	Forecast error: MAPE against WAPE

List of Abbreviations

Term	Expansion
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
CSV	Comma-Separated Values
DAX	Data Analysis Expressions
ETL / ELT	Extract-Transform-Load / Extract-Load-Transform
GBDT	Gradient Boosted Decision Trees
GMV	Gross Merchandise Value
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning

Term	Expansion
MLOps	Machine Learning Operations
NDCG	Normalised Discounted Cumulative Gain
OLTP	Online Transaction Processing
PSI	Population Stability Index
RLS	Row-Level Security
SLA	Service Level Agreement
UPI	Unified Payments Interface
WAPE	Weighted Absolute Percentage Error

Chapter 1: Introduction

1.1 Background

The operational deployment of machine learning has outpaced the discipline of maintaining it. Organisations that have successfully moved models from experimentation into production frequently discover that the harder problem is not building the model but knowing, at any given moment, whether it is still working.

This difficulty is structural rather than incidental. Conventional software communicates its own failure. A service that cannot reach its database returns an error; a process that exhausts memory terminates; a malformed request produces a status code that monitoring systems are designed to notice. Machine learning systems do not behave this way. A model whose training data has become unrepresentative of current conditions continues to accept requests, continues to execute, and continues to return outputs of the correct type and plausible magnitude. Nothing in the serving path distinguishes a well-calibrated prediction from a badly calibrated one.

Sculley et al. (2015) characterised this property as a form of technical debt particular to machine learning, noting that ML systems possess "all of the maintenance problems of traditional code plus an additional set of ML-specific issues," and that these issues are frequently invisible at the level of the system boundary. Subsequent work by Breck et al. (2017) proposed a production readiness rubric in which monitoring occupies a central position, and Paleyes et al. (2022), surveying deployment case studies across industries, found monitoring and maintenance to be among the most consistently underdeveloped stages of the machine learning lifecycle.

The commercial context in which this project is situated is the online home-services marketplace: a platform that connects customers requiring domestic work with independent tradespeople able to perform it. Such platforms are operationally demanding. They must forecast demand in order to position supply, rank available workers against incoming jobs, and price work whose true cost is not known until the worker arrives. Each of these is a machine learning problem, and each is executed continuously and at volume.

Bengaluru, the setting for this study, presents a demand environment with unusual and pronounced structure. The city experiences two distinct monsoon seasons — the south-west monsoon from June to September and the north-east monsoon from mid-October to late

November — and the seasonal calendar includes major festivals that materially alter consumption. These produce demand patterns that are predictable in aggregate but that shift the relationship between predictor variables and outcomes several times a year. A city with two monsoons is, consequently, a natural laboratory for studying model drift.

1.2 Problem Statement

The problem addressed by this project can be stated precisely.

Deployed machine learning models are conventionally monitored through accuracy metrics: error rates, classification scores, ranking measures. These metrics share a dependency that is rarely made explicit — they require ground truth. Computing the error of a demand forecast requires knowing what demand actually occurred, which is available only after the forecast horizon has elapsed. In practice this means accuracy monitoring is retrospective, and the interval between a system's failure and the observability of that failure is bounded below by the horizon of the prediction.

More seriously, accuracy metrics only move when the model's outputs become wrong. A model may be broken in the sense that it is no longer being maintained, no longer being retrained, and no longer receiving current data — and yet remain accurate, provided the relationship it learned continues to hold. Its accuracy communicates the stability of the environment, not the health of the pipeline. The two coincide only until the environment changes.

The consequence is a detection gap: the interval between the moment a pipeline ceases to function correctly and the moment any monitored metric reflects it. During that interval the organisation believes it is operating a maintained model. This project sets out to measure that gap under controlled conditions and to determine whether it can be closed.

1.3 Research Aim

To determine whether operational monitoring signals that do not depend on ground truth can detect a silent machine learning pipeline failure earlier than conventional accuracy metrics, and to quantify the difference in detection latency.

1.4 Research Objectives

2. To construct an end-to-end analytics and machine learning platform of

sufficient realism that monitoring behaviour observed within it is generalisable to production systems.

3. To develop and evaluate three machine learning services representative of marketplace operations: demand forecasting, worker ranking, and dynamic pricing.
4. To implement a monitoring layer capturing both accuracy metrics and operational metrics for eight deployed models.
5. To inject a pipeline failure at a known date and measure the detection latency of each class of monitoring signal.
6. To evaluate whether the accuracy metric conventionally applied to demand forecasting adequately represents the model's operational performance.
7. To derive practical monitoring recommendations from the observed results.

1.5 Research Questions

RQ1. Can an operational monitoring signal that requires no ground truth detect a silent pipeline failure earlier than an accuracy metric, and by what margin?

RQ2. Under what conditions does a stale model continue to perform acceptably, and what causes that performance to deteriorate?

RQ3. Does mean absolute percentage error, the conventional accuracy metric for demand forecasting, adequately characterise forecast quality in a setting containing zero-demand observations?

RQ4. What monitoring configuration would have detected the injected failure at the earliest opportunity, and what does it cost to implement?

1.6 Research Gap

The literature establishes clearly that monitoring is necessary. Sculley et al. (2015) identify its absence as a source of technical debt; Breck et al. (2017) include it in a production readiness rubric; Gama et al. (2014) provide a comprehensive treatment of drift detection

methods. Commercial and open-source tooling for drift monitoring is mature and widely available.

Three gaps remain, and this project addresses them.

First, detection latency is asserted rather than measured. The literature argues that monitoring detects problems earlier than the alternative, but rarely quantifies the interval. This is a consequence of the evidence base: production case studies report incidents after the fact, when the moment of failure is inferred rather than known. Without a known failure time, latency cannot be computed.

Second, the distinction between outcome-dependent and outcome-independent signals is under-examined. Drift detection literature concentrates on distributional comparison of inputs or outputs. Pipeline health signals — whether the retraining job succeeded, how old the training data is, whether the feature pipeline is delivering complete records — are treated as infrastructure concerns rather than model monitoring, despite being available at zero evaluation cost and with zero delay.

Third, forecast evaluation practice in operational settings is inherited rather than examined. MAPE is the default reporting metric for demand forecasting. Hyndman and Koehler (2006) demonstrated two decades ago that percentage errors are undefined when the actual value is zero and are asymmetric in their treatment of over- and under-prediction, yet the metric remains standard. Its behaviour in sparse operational settings, where zero-demand observations are common rather than exceptional, warrants direct measurement.

1.7 Structure of Research

Chapter 2 reviews the literature on machine learning technical debt, concept drift, monitoring practice, forecast evaluation, gradient boosting on tabular data, and dimensional modelling, concluding with the identified gap.

Chapter 3 describes the methodology: the design rationale for a synthetic data instrument, the construction of the dataset and warehouse, the development of three machine learning services, the failure injection protocol, and the validation strategy.

Chapter 4 presents and interprets the findings across five themes, discusses them in relation to the literature, and states their limitations.

Chapter 5 summarises the conclusions, links them explicitly to the research questions, offers recommendations, and reflects on the process.

1.8 Chapter Summary

This chapter has established that machine learning systems fail silently, that this property creates a measurable interval between failure and detection, and that the interval has not been directly quantified in the existing literature because production incidents do not come with known failure times. The research aim, objectives and questions have been stated, and the gap the project addresses has been specified in three parts: unmeasured detection latency, the neglect of outcome-independent monitoring signals, and unexamined forecast evaluation practice.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews the theoretical and empirical foundations of the study. It proceeds from the general characterisation of machine learning systems as maintenance liabilities, through the specific mechanisms by which deployed models degrade, to the practices proposed for detecting that degradation. It then examines forecast evaluation metrics, the modelling techniques appropriate to tabular operational data, and the dimensional modelling tradition on which the analytical layer of the artefact is founded. It concludes by locating the gap addressed by this project.

2.2 Technical Debt in Machine Learning Systems

The foundational treatment of maintenance burden in machine learning systems is Sculley et al. (2015), who applied the software engineering metaphor of technical debt to production machine learning. Their central argument is that machine learning systems accumulate a distinct class of liability arising from the entanglement of code, data and model, and that this liability is difficult to observe through the interfaces by which conventional software is monitored.

Several of their identified anti-patterns bear directly on this project.

Entanglement describes the property that changing any input feature changes the significance of all others, so that no component can be reasoned about in isolation. *Pipeline jungles* describe accreted data preparation chains whose failure modes are poorly understood. Most relevant here is their observation that machine learning systems frequently lack the plumbing that conventional systems take for granted — that the absence of straightforward monitoring is not an oversight in any particular system but a systematic characteristic of the field.

Breck et al. (2017) developed this into a practical instrument, proposing twenty-eight specific tests across data, model, infrastructure and monitoring. Their monitoring section explicitly includes tests for training data staleness and for the invariance of serving and training feature distributions, which supports the position taken in this project that pipeline health belongs within model monitoring rather than adjacent to it.

Paley et al. (2022) surveyed published deployment case studies and reported that monitoring and maintenance were among the least developed lifecycle stages in practice, with organisations frequently discovering degradation through downstream business impact rather than through instrumentation.

2.3 Concept Drift and Data Drift

The mechanism by which a deployed model loses accuracy without any change to its own parameters is described in the literature as drift. Gama et al. (2014) provide the standard taxonomy, distinguishing between changes in the input distribution and changes in the relationship between inputs and target.

The distinction matters for monitoring design. Input distribution change is observable immediately and without labels, by comparing the distribution of current inputs against the training distribution. Change in the input-output relationship is observable only through outcomes, and therefore only after those outcomes are known.

Gama et al. further distinguish drift by temporal profile — sudden, gradual, incremental and recurring. The seasonal structure examined in this project is best characterised as recurring drift: the monsoon arrives annually and shifts the demand regime in a manner that is predictable in timing though not necessarily in magnitude. Recurring drift is significant for the present study because it establishes that the exposure of a stale model is itself scheduled: if a model degrades when the regime changes, and the regime changes seasonally, then the consequences of a maintenance lapse are deferred to a date that can be anticipated in advance.

The population stability index is widely used in applied settings, particularly in credit risk, as a summary measure of distributional shift, with thresholds around 0.1 and 0.25 conventionally taken to indicate moderate and significant change respectively. These thresholds are heuristic rather than derived, a point returned to in Chapter 4.

2.4 Monitoring Practice for Deployed Models

Monitoring practice for deployed models divides into three broad categories.

Performance monitoring tracks the model's accuracy against ground truth. It is the most direct measure of whether the model is doing its job and the most delayed, since it requires outcomes to be observed.

Data monitoring tracks the statistical properties of inputs and outputs without reference to outcomes. It is available immediately and detects a necessary but not sufficient condition for degradation: inputs may shift without accuracy suffering, and accuracy may suffer without inputs shifting.

Operational monitoring tracks the system that produces predictions — latency, throughput, error rates, and the state of the training pipeline. This category is conventionally treated as infrastructure monitoring rather than model monitoring, and it is the category this project argues is under-used.

The distinction relevant to this study is not between these three categories as such, but between signals that require ground truth and those that do not. Training-data age falls into the operational category and requires nothing but a timestamp comparison; it is available continuously, at no computational cost, and with no delay. Its neglect in monitoring practice appears to be a matter of convention rather than of technical limitation.

2.5 Forecast Accuracy Measurement

The measurement of forecast accuracy has an extensive literature, and the limitations of percentage-based error metrics are long established.

Hyndman and Koehler (2006) provide the standard critique. They identify that MAPE is undefined when the actual value is zero, that it is asymmetric — penalising over-forecasting more heavily than under-forecasting of equivalent absolute magnitude — and that it can favour models that systematically under-predict. They propose the mean absolute scaled error as an alternative that remains defined across zero observations.

Despite this critique being two decades old, MAPE remains the default reporting metric in operational demand forecasting, appearing in service level agreements, vendor benchmarks and internal targets. The weighted absolute percentage error, which divides total absolute error by total actual volume, is sometimes reported alongside it and has the property of remaining defined when individual actuals are zero, since only the aggregate denominator must be non-zero.

The behaviour of these metrics diverges most sharply in sparse settings. Where demand is forecast at a fine grain — a single product in a single location on a single day — zero-demand observations are common. Under MAPE these observations are excluded from the average, since they cannot be computed. Under WAPE they contribute their absolute error

to the numerator while contributing nothing to the denominator. The two metrics therefore answer different questions, and the difference between them is informative in a way that is examined directly in Chapter 4.

Hyndman and Athanasopoulos (2021) provide the standard contemporary treatment of forecasting practice, including the observation that forecast evaluation should be conducted at the grain at which decisions are made, a principle applied in the evaluation protocol of this project.

2.6 Gradient Boosting for Tabular Prediction

The choice of model family for the three services was informed by the empirical literature on tabular prediction.

Gradient boosted decision trees, formalised in the implementations of Chen and Guestrin (2016) and Ke et al. (2017), remain the dominant approach for structured tabular data. Grinsztajn et al. (2022) examined the persistence of this dominance directly, finding that tree-based models continue to outperform neural architectures on typical tabular datasets, attributing this to the tendency of neural networks toward overly smooth decision boundaries, their sensitivity to uninformative features, and the rotational invariance that discards the meaning carried by individual columns.

For ranking problems, the LambdaMART family described by Burges (2010) provides the standard approach, optimising a ranking objective directly rather than optimising a pointwise prediction and sorting the results.

For prediction intervals rather than point estimates, quantile regression as introduced by Koenker and Bassett (1978) provides an approach that makes no distributional assumption, estimating specified quantiles of the conditional distribution directly.

One limitation of tree-based models is directly relevant to this project: a decision tree can only predict values present in its training data, and therefore cannot extrapolate a trend. Where a target grows monotonically over time, a tree model trained on earlier data will systematically under-predict later data. The standard remedy, adopted here, is to model a ratio to a known baseline rather than an absolute level.

2.7 Dimensional Modelling and the Semantic Layer

The analytical layer of the artefact follows the dimensional modelling tradition established by Kimball and Ross (2013). The star schema — a central fact table at a defined grain surrounded by denormalised dimension tables — is adopted for its query performance characteristics and, more importantly for this project, for its comprehensibility to non-specialist users.

Two Kimball principles are load-bearing in the present artefact. The first is that fact table grain must be declared before any other design decision, since every subsequent choice depends on it. The second is that of conformed dimensions: where multiple fact tables share a dimension, that dimension must be identical across them, permitting facts to be compared without direct relationships between fact tables.

The semantic layer — the collection of named, reusable calculations defined over the dimensional model — is the mechanism by which business definitions are centralised. Where a definition such as "active customer" is expressed once in a semantic layer rather than repeatedly in individual reports, divergence between reports is prevented structurally rather than by convention.

2.8 Synthetic Data as an Evaluation Instrument

The use of synthetic data in this study is methodological rather than substitutive, and the distinction requires justification.

Synthetic data is commonly employed where real data is unavailable, restricted by privacy regulation, or insufficient in volume. In such applications the synthetic dataset is a proxy, and its value depends on its fidelity to the real distribution it replaces.

The use here is different. The research question concerns detection latency: the interval between a failure occurring and a monitoring signal responding to it. Measuring an interval requires both endpoints to be known. In production data the second endpoint is observable but the first is not — incident post-mortems infer the moment of failure, frequently incorrectly, and always after the fact.

A synthetic instrument permits the first endpoint to be specified rather than inferred. The failure is placed at a chosen date; detection is then measured against it exactly. This is the same logic by which a fault is injected into a system under test rather than waited for. The validity of the resulting measurement depends not on the dataset resembling any particular

real marketplace, but on the failure mechanism and the monitoring signals behaving as they would in a real system.

2.9 Research Gap Identified in the Literature

The review supports the gap stated in Section 1.6. Monitoring is established as necessary and its methods are well developed, but detection latency is asserted rather than measured, because the evidence base consists of production incidents whose failure times are unknown. Outcome-independent pipeline signals are recognised in production readiness rubrics yet are not treated as first-class monitoring instruments. Forecast evaluation practice continues to rely on a metric whose limitations were documented two decades ago, without direct measurement of the consequences in sparse operational settings.

Chapter 3: Methodology

3.1 Research Design

The study adopts a **design science** approach, in which knowledge is produced through the construction and evaluation of an artefact. The artefact is a complete analytics and machine learning platform; the evaluation is a controlled experiment conducted within it.

The design is quantitative and experimental. A failure is injected at a known time and detection latency is measured for two classes of monitoring signal. The independent variable is the class of signal; the dependent variable is the interval between failure injection and signal divergence.

The requirement for a known failure time determines the choice of a synthetic data instrument, for the reasons set out in Section 2.8.

Determinism was treated as a design requirement rather than a convenience. The entire dataset is generated from a single fixed pseudo-random seed, so that regeneration produces a byte-identical result. Every figure reported in Chapter 4 is therefore reproducible, and any reader may verify it by executing the generation and validation programs.

3.2 Data Generation

A dataset was generated representing twenty months of operation — 1 January 2025 to 31 August 2026, 608 consecutive days — of a home-services marketplace in Bengaluru offering 37 services across eight trade categories in 20 localities.

3.2.1 Structural realism

Certain elements were taken from reality rather than generated. The 20 localities are actual Bengaluru neighbourhoods with correct postal codes, administrative zones and approximate coordinates. Both monsoon seasons are modelled. The festival calendar and the Indian fiscal year, running April to March, are represented. Price points were benchmarked against prevailing rates in the city, and the payment mix reflects the dominance of the Unified Payments Interface in Indian consumer transactions.

3.2.2 Behavioural modelling

Relationships in the data were generated as causal chains rather than sampled correlations. Rainfall suppresses exterior painting and increases plumbing demand. Heavy-rain days raise

cancellation rates, which reduce completion rates, which reduce revenue. Daily volume above the trailing thirty-day mean lengthens time-to-assignment, which increases service level breaches, which reduces customer ratings.

Seasonal multipliers are declared as configuration and composed multiplicatively where windows overlap. Exterior painting during the overlap of the Diwali festival window and the north-east monsoon therefore carries a multiplier of $2.4 \times 0.65 = 1.56$: demand rises for the festival but rain continues to restrain it. This value is asserted by a unit test rather than by a comment.

3.2.3 Consistency by construction

Two design decisions eliminate categories of inconsistency that synthetic datasets commonly exhibit.

Bookings are assigned to technicians by consuming actual capacity slots from a capacity calendar generated in advance. The daily capacity fact therefore reconciles exactly to the booking fact, rather than approximately.

The conversion funnel is constructed backwards from realised bookings. Quotes are derived from bookings, leads from quotes, and searches from leads, each with tier-dependent conversion rates. Monotonicity of the funnel holds by construction and cannot be violated.

3.3 Warehouse and Semantic Layer Construction

A dimensional model of eleven tables was constructed: six dimensions and five facts, arranged as a star schema with conformed date, area and service dimensions. No relationships exist between fact tables.

A source-system schema was specified in PostgreSQL representing the transactional database from which such data would originate, including an append-only booking event log and a prediction store. A staging layer performs renaming, typing and flattening; a mart layer holds business definitions. Business logic is expressed once, in the mart layer.

A semantic layer of 119 measures was implemented in DAX, organised into eight display folders, each measure carrying an explicit format string. The measures were generated from a single declarative source specification, so that the human-readable library and the deployment script cannot diverge.

Two dashboard implementations were produced over the same semantic definitions: a Power BI report and a Streamlit web application. Producing both permitted a practical comparison of the two approaches and provided a cross-check on the measure definitions, since divergence between the two implementations indicates an error in one of them. One such divergence was detected and is reported in Section 4.3.

3.4 Model Development

Three services were developed, trained and evaluated.

3.4.1 Demand forecasting

The forecasting model predicts total job volume over the following seven days for each area and service category. The training data comprises a dense daily panel of 97,280 observations.

Features comprise lagged volumes at 1, 2, 3, 7, 14, 21 and 28 days; rolling means and maxima over 7, 14 and 28 days; a momentum ratio; calendar features; and the configured seasonal multiplier both for the observation date and averaged across the forecast horizon. All lag and rolling features are shifted by one day to prevent leakage.

Evaluation used a strictly forward temporal split, training on observations prior to 1 May 2026 and validating on subsequent observations.

An initial specification predicting absolute volume produced a bias of -39.6% . The cause was diagnosed as the inability of tree ensembles to extrapolate: demand grows approximately 4.5-fold across the study window, and validation volumes consequently exceeded any value present in training. The specification was revised to a Poisson objective with a logarithmic exposure offset equal to seven times the trailing daily mean, so that the model predicts a multiplier on a known baseline. Bias fell to -6.8% .

3.4.2 Technician ranking

The ranking model orders available technicians for an incoming job. Since the fact table records only the technician selected, negative examples were constructed by sampling from technicians who were genuinely available on the same day. A LambdaMART objective was trained over 14,000 query groups of eight candidates each. Splitting was performed by query group rather than by row, since splitting within a group would place a job's selected technician in training and its rejected candidates in validation, rendering the task trivial.

3.4.3 Dynamic pricing

The pricing model comprises three quantile regressors at the 10th, 50th and 90th percentiles, producing a price band rather than a point estimate, together with a binary classifier estimating acceptance probability. The quoted amount is excluded from the band model's features, since predicting the final price from the quote would be circular.

3.5 Failure Injection and Measurement Protocol

The experimental manipulation is the injection of a pipeline failure at a known date, together with an exogenous regime change at a second known date.

Date	Event
15 March 2026	The scheduled weekly retraining job ceases to succeed. No alert is raised; the model continues to serve predictions from its existing version.
1 June 2026	The south-west monsoon begins. Plumbing demand approximately doubles; exterior painting demand falls.
15 June 2026	Population stability index crosses the conventional 0.25 alert threshold.
18 June 2026	Forecast error reaches a degraded plateau.
20 July 2026	Retraining is restored. Model version increments; training-data age resets.

Two signal classes were monitored daily throughout:

Operational signals, requiring no ground truth: training-data age, feature null rate, prediction volume, serving latency.

Accuracy signals, requiring ground truth: mean absolute percentage error, weighted absolute percentage error, forecast bias.

Detection latency for each signal is defined as the interval between 15 March 2026 and the first date on which the signal departs materially from its pre-failure behaviour.

3.6 Validation Strategy

Three independent layers of verification were implemented.

Data integrity. Sixteen automated checks verify referential integrity across every fact and dimension, contiguity of the date dimension, money rules including the identity between platform revenue and commission applied to the final amount, the confinement of

completion-only columns to completed bookings, chronological consistency, funnel monotonicity, and capacity reconciliation. The build fails if any check fails.

Unit and integration testing. 109 automated tests cover the seasonality window logic, the feature builders including explicit temporal leakage checks, the service contracts, and the dashboard pages.

Cross-implementation comparison. Headline figures produced by the Streamlit application were compared against those produced by the validation programme directly from the source files.

Each layer detected defects that the others did not, and these are reported in Section 4.3 as a finding in their own right.

3.7 Ethical Considerations

No human participants were involved and no personal data was processed. All records are synthetic, generated from a fixed seed, and no real individual, organisation or transaction is represented. The fictional nature of the dataset is stated prominently in the artefact's documentation and in its user interface, so that no reader may mistake generated figures for empirical observations of a real business.

The source-system schema was designed to exclude personally identifying attributes from the analytical layer, demonstrating the data minimisation principle: the warehouse receives a customer key and behavioural attributes, and no name, telephone number or email address.

Reporting honesty was treated as an ethical requirement rather than a stylistic preference. Two of the three models produced results that do not support deployment, and both are reported with the measured figure, the diagnosis and the recommended remedy rather than being omitted or tuned until presentable.

3.8 Chapter Summary

This chapter has described a design science study in which a complete analytics and machine learning platform was constructed as the instrument for a controlled experiment. The synthetic data instrument was justified by the requirement for a known failure time. The dataset, warehouse, semantic layer and three machine learning services have been described, together with the failure injection protocol, the measurement definitions, the three-layer validation strategy and the ethical position.

Chapter 4: Discussion and Analysis

4.1 Introduction

This chapter presents the results. Section 4.2 describes the dataset produced. Section 4.3 reports findings across five themes. Section 4.4 interprets them, Section 4.5 discusses them in relation to the literature, and Section 4.6 states their limitations.

4.2 Overview of the Dataset

Table	Rows	Grain
dim_date	608	One day
dim_service	37	One service
dim_area	20	One locality
dim_professional	850	One technician
dim_customer	24,000	One customer
dim_model	8	One deployed model
fact_bookings	57,973	One booking
fact_pro_capacity	324,435	One technician-day
fact_leads	74,286	Day $\times$ area $\times$ service
fact_model_metrics	3,481	Day $\times$ model
fact_forecast_accuracy	47,117	Day $\times$ area $\times$ category

Of 57,973 bookings, 46,394 were completed, a rate of 80.0%. Gross merchandise value totals ₹124,526,480 with platform revenue of ₹22,536,667, a blended take rate of 18.1% and an average order value of ₹2,684. Monthly completed volume grows from 910 in January 2025 to 4,134 in August 2026.

All sixteen integrity checks pass, as do 109 automated tests.

4.3 Findings

Theme 1: Detection latency differs by two and a half months between signal classes

This is the study's principal result and the direct answer to RQ1.

Signal	Requires ground truth	First divergence	Latency
Training-data age	No	16 March 2026	1 day
Feature null rate	No	1 June 2026	78 days

Signal	Requires ground truth	First divergence	Latency
Population stability index	No	15 June 2026	92 days
Mean absolute percentage error	Yes	18 June 2026	95 days

Training-data age follows a seven-day sawtooth under normal weekly retraining, rising to seven and resetting. From 16 March 2026 the reset does not occur and the value climbs monotonically, reaching a maximum of 126 days before retraining is restored on 20 July 2026. The departure from normal behaviour is unambiguous on the first day it occurs.

Forecast error remains statistically indistinguishable from its pre-failure level throughout March, April and May. The pre-incident mean is 8.93%; the model remained within its 12% target for the entire period during which it was unmaintained.

The detection gap is therefore 94 days.

Theme 2: A stale model degrades only when the environment changes

The interval between failure and accuracy degradation is explained by the absence of environmental change during it.

Between March and June the relationship between the seasonal features and demand remained stable. A model fitted in March described June's conditions adequately because June's conditions resembled March's. Accuracy was measuring the stability of the environment, not the health of the pipeline.

The south-west monsoon on 1 June altered the relationship. Plumbing volume rose from 471 jobs in May 2026 to 1,026 in August; exterior painting fell from 209 to

8. Forecast error rose to a plateau of 19.31%, against a pre-incident baseline

of 8.93%. Following restoration of retraining on 20 July, error recovered to 10.00% within four days.

The staleness was the cause; the monsoon was only the trigger.

Theme 3: MAPE conceals a failure mode it cannot represent

The forecasting model's mean absolute percentage error across the full period is

10.4%. Its weighted absolute percentage error over the same predictions is

37.2%.

Decomposition attributes the entire divergence to a single population.

Cell type	Cells	Absolute error	Actual jobs
Actual demand present	36,113	5,871	55,654
Actual demand absent	11,004	14,806	0

11,004 forecast cells contain a prediction of demand where none occurred, and these cells carry 14,806 jobs of predicted volume — 26.6% of total realised volume. Excluding them, WAPE is 10.5%, in close agreement with MAPE.

MAPE cannot represent these cells. Percentage error requires division by the actual value; where the actual is zero the quantity is undefined and the observation is excluded from the average. The metric is not merely insensitive to this failure mode but mathematically incapable of expressing it.

The operational interpretation is direct: each such cell represents supply potentially positioned against demand that did not materialise.

Theme 4: Model quality assessed honestly, including where it is poor

Demand forecasting. At the cell grain, held-out MAPE is 47.7%. A parametric bootstrap in which actuals are simulated from a perfectly calibrated model establishes an irreducible noise floor of 44.2%, since the mean target at that grain is approximately seven jobs and small counts are dominated by Poisson variance. The model sits 3.5 percentage points above a bound no model can beat. At the grain at which supply decisions are made — an area, or a category across the city — held-out MAPE is 13.7% and 14.7% respectively.

Technician ranking. NDCG@5 is 0.954 against a target of 0.82. **This result is optimistic and is not evidence of a well-performing ranker.** The data generator assigns jobs using a known deterministic function of tier, distance and category match — the same features the model receives. The model recovers a specified process rather than learning a human one. Recall@5 of 1.000 across eight candidates is close to uninformative; NDCG@1 of 0.883 is the more meaningful figure and remains flattered.

Dynamic pricing. The price band is sound: the 10th-to-90th percentile interval contains 78.2% of realised amounts against a nominal 80%, with no quantile crossings, and median absolute error of ₹361 on ticket sizes spanning ₹450 to ₹45,000.

The acceptance classifier achieves AUC 0.527 against a base rate of 79.4%, and

is not fit for deployment. Operational load features were added specifically to improve it; AUC moved from 0.530 to 0.527. The diagnosis is that booking completion in this dataset is driven principally by a per-day rainfall variable absent from the feature set, and no available feature proxies it adequately. The remedy is a weather data feed, not further tuning.

Theme 5: Automated verification detected defects that review did not

Six defects were detected by automated checks rather than by inspection.

Detected by	Defect
Integrity checks	Six technicians assigned work after their recorded departure date, caused by the same tenure calculation being implemented twice with differing edge-case handling
Service tests	Training-serving skew: a feature added to the training frame but not to the single-row inference frame
Dashboard tests	A join collision in which a column present on both fact and dimension was silently renamed, causing every downstream reference to fail
Cross-implementation comparison	A threshold computed across bookings in one implementation and across days in the other, producing 34.8% where the reference produced 32.5%
Profiling	Peak memory of 603 MB against a 1 GB deployment ceiling, of which 141 MB was low-cardinality text stored per row
Bias measurement	Systematic −39.6% forecast bias from tree extrapolation failure

The training-serving skew is notable: training completed successfully and reported valid metrics while the inference path raised an exception. No review of the training code would have revealed it.

Supplementary findings

Three findings emerged that were not sought by the research questions.

Seasonal demand is a reallocation problem. Technician utilisation during monsoon rises for pest control (+59%), plumbing (+57%) and electrical work (+38%) while falling for painting (−12%) and air-conditioning (−25%). Aggregate utilisation is 17.0%. Idle capacity exists concurrently with strained capacity in the same months.

Acquisition channel predicts value more strongly than volume. Referred customers repeat at 79.0% with mean lifetime value ₹7,793; paid-social customers repeat at 39.9% with mean lifetime value ₹3,506 — a 2.2-fold difference — while the latter channel supplies the second-largest customer count.

Service level failure concentrates on predictable days. Partitioning completed jobs by whether daily volume exceeded its trailing thirty-day mean, breach rates are 32.5% on the busiest quintile against 14.1% otherwise; mean time to assignment rises from 11.4 to 19.7 minutes and mean rating falls from 4.32 to 4.04.

4.4 Interpretation of Findings

4.4.1 The significance of outcome independence

The 94-day gap is not explained by the sophistication of either signal. Training- data age is arithmetically trivial — a subtraction of two dates. Its advantage is structural: it observes the pipeline rather than the predictions, and the pipeline broke first.

This suggests monitoring signals should be classified by what they observe rather than by how they are computed. A signal observing the process can respond when the process fails. A signal observing outputs can respond only when outputs become wrong, which may be considerably later or, if the environment remains stable, never.

4.4.2 Accuracy as a lagging indicator

Theme 2 indicates that accuracy metrics measure the conjunction of two conditions: that the model was appropriately fitted, and that the environment has not since changed. A stable accuracy reading is consistent with a well-maintained model in a stable environment, and equally consistent with an abandoned model in a stable environment. The reading alone does not distinguish them.

This has a practical consequence for recurring drift. If the environment changes seasonally and a maintenance lapse occurs during a stable period, the consequence is deferred to the next regime change. The organisation experiences a sudden degradation whose proximate cause — the monsoon — is visible, and whose actual cause — a maintenance failure three months earlier — is not.

4.4.3 Metric selection as a design decision

Theme 3 shows a single model receiving assessments of 10.4% and 37.2% error depending on metric choice. Both computations are correct.

The relevant property is which failure modes each metric can express. MAPE cannot express prediction into zero-demand cells. Reporting it alone renders a substantial failure mode structurally invisible, irrespective of how carefully the number is monitored.

4.4.4 Honest reporting as a methodological requirement

Theme 4 includes two results that do not support deployment. Reporting them is methodologically necessary rather than merely commendable.

The ranking result is the more instructive. A model scoring 0.954 against a target of 0.82 would ordinarily be reported as a success. It is an artefact of the evaluation instrument: the generator specifies assignment as a function of the features the model receives, so high performance measures the recoverability of a specified process. Reporting the figure without that caveat would misrepresent the model's expected production behaviour.

4.4.5 Verification layers as independent instruments

Theme 5 indicates that each verification layer detected a distinct defect class, and that no layer would have detected the full set. This supports a layered verification strategy over reliance on any single method.

4.5 Discussion of Findings

4.5.1 Relation to the technical debt literature

The results provide direct empirical support for Sculley et al. (2015). The injected failure is precisely the class of liability they describe: invisible at the system boundary, accumulating without symptom, and surfacing later through a mechanism apparently unrelated to its cause.

The finding extends their argument by quantifying the interval. Their treatment establishes that such debt exists; this study measures 94 days of it under controlled conditions.

4.5.2 Relation to drift detection literature

The results complement Gama et al. (2014) while suggesting a boundary to the drift framing. In this study drift detection performed as the literature predicts: population stability index crossed its threshold and identified the degradation. But it did so 92 days after the failure

and only 3 days before the accuracy metric, because drift detection also observes data rather than pipeline.

The implication is not that drift detection is inadequate but that it addresses a different failure class. Drift detection identifies environmental change; pipeline monitoring identifies maintenance failure. A system experiencing both requires both.

The conventional 0.25 threshold performed adequately here, though the value is heuristic. Its adequacy in this instance should not be generalised.

4.5.3 Relation to forecast evaluation literature

Theme 3 provides a contemporary operational illustration of the critique advanced by Hyndman and Koehler (2006). Their argument that percentage errors are undefined at zero actuals is not novel; what this study adds is a measurement of the magnitude at stake in a realistic sparse setting — 26.6% of total volume forecast into cells where nothing occurred, entirely invisible to the reported metric.

The persistence of MAPE two decades after its documented limitations suggests the obstacle is conventional rather than technical.

4.5.4 Implications for practice

Three implications follow.

Monitor the pipeline, not only the predictions. At least one signal should be independent of ground truth. Training-data age is the cheapest available and would have detected the failure on the day it occurred.

Report a companion metric to any percentage error. WAPE, MASE or an explicit count of zero-actual cells. The divergence between metrics is itself diagnostic.

Assume degradation is deferred, not absent. A model that has not degraded may be unmaintained in a stable environment. Where the environment changes on a known seasonal cycle, the cost of a maintenance lapse is incurred at the next cycle.

4.6 Limitations of Findings

4.6.1 Synthetic data

The failure mechanism was specified by the researcher. The finding that training-data age responds on the day of failure is partly definitional: the signal was constructed to reflect the

pipeline state that was altered. What is not definitional, and constitutes the finding, is the 94-day interval before accuracy responded, which follows from the interaction between the injected failure and the independently specified seasonal structure.

Absolute values are properties of this dataset. The 94-day interval would differ under a different seasonal calendar or a different failure date.

4.6.2 Single failure mode

One failure mode was examined: cessation of retraining. Other modes — schema change, upstream data corruption, feature computation error, label delay — were not, and may exhibit different detection profiles.

4.6.3 Model evaluation constraints

The ranking evaluation is compromised by the recoverability of the generative process, as reported. The pricing acceptance classifier is limited by the absence of the driving variable from the feature set. Both are stated rather than concealed, but both limit what may be concluded about model quality as distinct from monitoring behaviour.

4.6.4 Single-domain scope

One domain in one city was examined. The seasonal structure of Bengaluru is unusually pronounced, and may accentuate the divergence between pipeline and accuracy signals relative to a more stable environment.

4.7 Chapter Summary

This chapter reported five themes. Detection latency differed by 94 days between outcome-independent and outcome-dependent signals. A stale model degraded only on environmental change. MAPE of 10.4% concealed WAPE of 37.2%, attributable entirely to 11,004 zero-actual cells. Model quality was reported including two results that do not support deployment. Automated verification detected six defects that review did not. Findings were interpreted, discussed against the literature, and their limitations stated.

Chapter 5: Conclusion and Recommendations

5.1 Introduction

This chapter summarises the conclusions, links them to the research questions, states the study's limitations, offers recommendations for practice and further research, and reflects on the process.

5.2 Summary of Key Findings

Operational monitoring detects pipeline failure 94 days earlier than accuracy monitoring. Training-data age diverged one day after the injected failure; forecast error diverged 95 days after it.

Stable accuracy does not indicate a healthy pipeline. It indicates that the model was appropriately fitted and that the environment has not since changed. A maintenance lapse during a stable period defers its cost to the next regime change.

Percentage error metrics conceal failure modes they cannot express. MAPE of 10.4% and WAPE of 37.2% described the same predictions; the divergence was entirely attributable to 11,004 cells forecasting 14,806 jobs where none occurred, 26.6% of realised volume.

Layered automated verification detects defects that review does not. Six defects were found by six different mechanisms, including training-serving skew invisible to inspection of the training code.

5.3 Limitations of Study

The findings derive from a synthetic instrument with a researcher-specified failure. Absolute intervals are properties of this dataset. One failure mode was examined. Two of three models carry evaluation constraints that are reported but that limit conclusions about model quality. The study covers one domain in one city with an atypically pronounced seasonal structure.

5.4 Link to Research Questions and Objectives

RQ1 — Can an outcome-independent signal detect silent failure earlier, and by what margin? Answered affirmatively. Training-data age detected the failure at one day; MAPE at 95 days. The margin is 94 days.

RQ2 — Under what conditions does a stale model continue to perform, and what causes deterioration? Answered in Theme 2. A stale model performs while the learned relationship holds. Deterioration was caused by the seasonal regime change of 1 June 2026, which the staleness rendered the model unable to accommodate.

RQ3 — Does MAPE adequately characterise forecast quality where zero-demand observations occur? Answered negatively. MAPE excluded 11,004 of 47,117 cells by mathematical necessity, concealing 26.6% of realised volume in phantom predictions.

RQ4 — What configuration would have detected the failure earliest, and what does it cost? A threshold alert on training-data age exceeding twice the retraining cadence. It requires one timestamp comparison, no ground truth and no model evaluation, and would have fired on 16 March 2026.

All six objectives stated in Section 1.4 were met. The artefact was constructed and validated; three services were developed and evaluated; monitoring was implemented across eight models; the failure was injected and latency measured; forecast metric adequacy was examined; and recommendations were derived.

5.5 Recommendations

5.5.1 For practice

9. **Alert on training-data age for every deployed model**, at a threshold of

approximately twice the intended retraining cadence. This is the study's principal recommendation. It is arithmetically trivial and would have detected the failure on the day it occurred.

10. **Alert on the absence of retraining success, not the presence of failure.** A

job that stops running emits no error. Monitoring should treat silence as a signal.

11. **Report a companion metric alongside any percentage error**, and count

zero-actual cells explicitly.

12. **Evaluate forecasts at the grain of the decision.** Cell-grain error was 47.7%

against a noise floor of 44.2%; area-grain error was 13.7%. Only the latter informs a staffing decision.

13. Treat pipeline health as model monitoring, not as infrastructure

monitoring. The signals belong on the same dashboard as the accuracy metrics.

5.5.2 For further research

14. Extend the protocol to additional failure modes — schema change, upstream

corruption, feature computation error, label delay — to determine whether the latency advantage generalises.

15. Vary the interval between failure and regime change systematically, to

characterise the relationship between environmental stability and detection delay.

16. Replicate against production data where a failure time is independently

known from deployment logs, providing external validation.

17. Examine threshold selection empirically. The 0.25 population stability

index threshold is heuristic; its false positive and negative rates in operational settings are not well established.

5.6 Reflections in First Person

The most useful thing I learned in this project came from a result I did not expect and initially assumed was a mistake.

When I first computed the forecasting model's error I obtained 10.4% by one metric and 37.2% by another over identical predictions. My first assumption was a bug in my own code. It took some time to accept that both figures were correct and that the difference was the finding rather than an obstacle to it. That MAPE excludes zero-actual observations is elementary and I knew it; I had not appreciated that in a sparse operational setting this exclusion could hide a quarter of total volume.

The second thing I learned concerns verification. I wrote sixteen integrity checks expecting them to confirm work I had already done correctly. They found six technicians assigned jobs after their departure dates, caused by my having implemented the same tenure calculation twice with different edge-case handling. Later, service tests found a feature I had added to training and forgotten to add to inference — a defect no reading of the training code would have revealed.

I also had to decide how to report two models that did not work. The ranker scored well above target for reasons that made the score meaningless, and the pricing acceptance classifier performed at approximately chance. The straightforward option was to report the good number and omit the caveat. I concluded that a result reported without its caveat is not a result, and that a model card containing only favourable outcomes serves no purpose. I found the discipline of writing "this component is not fit to ship" harder than expected and more valuable than the components that worked.

Finally, building the same dashboard twice — once in Power BI and once in Streamlit — was not planned as a verification method but became one. The two implementations disagreed about one figure, and the disagreement was a genuine error in one of them. Redundancy proved more effective at finding mistakes than care.

References

Note for submission: every reference below should be verified against the original publication before submission, and the citation style adjusted to the convention required by the programme.

Breck, E., Cai, S., Nielsen, E., Salib, M. and Sculley, D. (2017) 'The ML Test Score: A Rubric for ML Production Readiness and Technical Debt Reduction',

Proceedings of the IEEE International Conference on Big Data, pp. 1123–1132.

Burges, C.J.C. (2010) *From RankNet to LambdaRank to LambdaMART: An Overview*. Microsoft Research Technical Report MSR-TR-2010-82.

Chen, T. and Guestrin, C. (2016) 'XGBoost: A Scalable Tree Boosting System',

Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794.

Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M. and Bouchachia, A. (2014) 'A Survey on Concept Drift Adaptation', *ACM Computing Surveys*, 46(4), pp. 1–37.

Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022) 'Why do tree-based models still outperform deep learning on typical tabular data?', *Advances in Neural Information Processing Systems* 35.

Hyndman, R.J. and Athanasopoulos, G. (2021) *Forecasting: Principles and Practice*. 3rd edn. Melbourne: OTexts.

Hyndman, R.J. and Koehler, A.B. (2006) 'Another look at measures of forecast accuracy', *International Journal of Forecasting*, 22(4), pp. 679–688.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. (2017) 'LightGBM: A Highly Efficient Gradient Boosting Decision Tree', *Advances in Neural Information Processing Systems* 30, pp. 3146–3154.

Khanuja, S., Bansal, D., Mehtani, S. et al. (2021) *MuRIL: Multilingual Representations for Indian Languages*. arXiv preprint arXiv:2103.10730.

Kimball, R. and Ross, M. (2013) *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*. 3rd edn. Indianapolis: Wiley.

- Koenker, R. and Bassett, G. (1978) 'Regression Quantiles', *Econometrica*, 46(1), pp. 33–50.
- Paleyes, A., Urma, R.-G. and Lawrence, N.D. (2022) 'Challenges in Deploying Machine Learning: A Survey of Case Studies', *ACM Computing Surveys*, 55(6), pp. 1–29.
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F. and Dennison, D. (2015) 'Hidden Technical Debt in Machine Learning Systems', *Advances in Neural Information Processing Systems* 28, pp. 2503–2511.

Appendix 1: Declaration of Authenticity

I declare that this work is my own, that all sources have been acknowledged, and that the artefact described was constructed by me for the purpose of this study. The dataset analysed is synthetic and was generated by the programs included in the accompanying repository; it does not represent any real organisation, individual or transaction, and this is stated in the artefact's documentation and user interface.

Signed: _____ *Date:* _____

Appendix 2: Artefact Inventory

Repository: github.com/ABHISHEKPRASAD16/seek-my-service-analytics Deployed application: [seek-my-service-analytics.streamlit.app](#)

Component	Location	Description
Data generator	generator/	Configuration, seasonality logic, generation program
Generated dataset	data/	Eleven CSV files, deterministic from seed 20260819
Validation	validate.py	Sixteen integrity checks
Source schema	sql/01_source_schema.sql	PostgreSQL OLTP schema
Staging layer	sql/02_staging_views.sql	Renaming, typing, flattening
Dimensional build	sql/03_star_schema.sql	Mart layer with business definitions
Semantic layer	powerbi/measures.dax	119 DAX measures
Deployment scripts	powerbi/*.csx	Tabular Editor automation
ML services	ml/	Three FastAPI services, shared features, training
Dashboard	app/, streamlit_app.py	Five-page web application
Tests	tests/	109 automated tests
Documentation	docs/	Data dictionary, model cards, architecture

Appendix 3: Validation Output

All sixteen checks pass on the submitted dataset.

#	Check
1	Referential integrity across every fact and dimension
2	Date dimension contiguous across the full range
3	No negative amounts; final amount within quoted plus discount
4	Completion-only columns populated only for completed bookings
5	Signup precedes first booking; join date precedes first job
6	Rating and sentiment blank together, populated together
7	Funnel monotonic and reconciling to the booking fact
8	Capacity reconciling to bookings per technician per day
9	Row counts per table
10	Monthly booking counts showing the growth curve
11	Monsoon, summer and festival signatures present
12	Drift incident and recovery visible in the error series
13	Service level breach worse on high-strain days
14	Referral and organic acquisition out-repeating paid social
15	Payment mix dominated by UPI
16	Total gross merchandise value and platform revenue

Declaration of Authenticity

I hereby declare that this work is my own and that all sources used have been acknowledged. The artefact described in this report was designed, implemented and evaluated by me for the purpose of this study.

The dataset analysed is synthetic and was generated by the programs contained in the accompanying repository. It does not represent any real organisation, individual or transaction, and this is stated in the artefact's documentation and user interface.

I further declare that this work has not been submitted for any other degree or qualification.

Abhishek Prasad

Abhishek Prasad

Berlin, 01.10.2026